

Data Collection for AI

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 9: AI Security Lifecycle

1. Why Data Is the Foundation of Security AI

Every AI model is only as good as the data it learns from. This principle — often expressed as 'garbage in, garbage out' — is especially consequential in security, where AI-driven decisions may block or allow access, flag or ignore threats, and escalate or dismiss incidents. A model trained on incomplete, biased, or tampered data produces predictions that are unreliable in precisely the scenarios where reliability matters most.

Unlike general-purpose AI applications, security AI faces an adversarial environment. Attackers actively seek to understand and manipulate the systems defending against them. This means data quality is not just a technical concern but a security control in its own right. Protecting the integrity of training and inference data is as important as protecting the model itself. Data collection is therefore the first — and arguably the most consequential — phase of the AI Security Lifecycle.

2. Identifying Data Requirements

Before collecting anything, define exactly what data the AI needs to achieve its objective. This sounds straightforward but is frequently skipped in favour of collecting whatever data is available. The result is datasets that are large but not relevant, which wastes storage, compute, and preprocessing effort while adding noise to the model's inputs.

A practical technique is to ask: what information would an experienced security analyst examine before making this decision? If the AI is detecting malware, analysts look at file behaviour, API calls, strings, and entropy. If the AI is identifying insider threats, analysts look at access patterns, data movement, and deviation from baseline behaviour. Those analytical signals become the features the AI needs — and identifying them up front tells you exactly which data sources to prioritise.

AI Security Use Case	Required Data Types	Key Signal Examples
Malware detection	File samples (benign and malicious), execution traces	API call sequences, PE headers, string entropy, packed sections
User behaviour analytics (UBA)	Authentication logs, access logs, HR data (role, tenure)	Login time deviation, unusual resource access, privilege escalation
Network intrusion detection	NetFlow records, packet captures, DNS logs	Connection volume, beacon intervals, C2 callback patterns
Threat intelligence assistant	Threat reports, IOC feeds, vulnerability databases	CVE descriptions, MITRE ATT&CK mappings, actor TTPs
Phishing detection	Email headers, body text, URL features, sender reputation	Spoofed display names, lookalike domains, urgency language
Cloud misconfiguration detection	Cloud API logs, configuration snapshots, policy definitions	Public S3 buckets, overprivileged roles, disabled MFA

3. Data Trustworthiness and Poisoning Risk

Collecting data is insufficient if that data cannot be trusted. In a security context, data trustworthiness requires evaluating four properties: accuracy (does the data correctly describe what happened?), completeness (are all relevant events captured?), integrity (has the data been altered?), and provenance (can the origin of every record be verified?).

Data poisoning is an active attack technique in which an adversary deliberately introduces false or misleading records into a training dataset. A poisoned malware classifier might be trained to ignore a specific obfuscation technique used by the attacker. A poisoned UBA model might have its baseline shifted so that the attacker's reconnaissance activity appears normal. Defences against poisoning include strict access controls on data pipelines, cryptographic validation of records at ingestion, anomaly detection on the data itself before training, and maintaining a separate held-out validation set whose integrity is independently verified.

Key Principle: Trust in a model cannot exceed trust in its training data. Data integrity checks are security controls, not optional quality improvements.

4. Real-World Incident: Supply Chain Attack Demonstrating Data Integrity Risk

Real-World Incident — SolarWinds / SUNBURST (2020): Nation-state attackers (attributed to SVR / Cozy Bear) compromised SolarWinds' build pipeline and inserted malicious code into the Orion software update. Organisations that trusted the update implicitly — because it came from a trusted vendor with a valid signature — deployed malware to their own networks. Applied to AI security: if an attacker can compromise the data pipeline feeding a security AI system, every model trained on that pipeline inherits the corruption. The SolarWinds attack illustrates why supply chain integrity and data provenance verification are not optional hygiene measures but active security requirements in any AI system that ingests external data.

The lesson for AI security practitioners is that data authenticity cannot be assumed — it must be verified. Digital signatures on data feeds, cryptographic hashes on datasets at rest, and chain-of-custody records for training datasets are practical controls that reduce the risk of supply-chain-style data compromise.

5. Secure Data Collection Methods

Collecting sensitive security data creates an attractive target. Authentication logs, endpoint telemetry, network traffic captures, and threat intelligence contain information that attackers would find valuable for planning future operations or understanding an organisation's detection capabilities. The collection pipeline must therefore apply the same rigour to protecting the data in transit and at rest as it applies to protecting production systems.

- Encryption in transit — all data flowing from source systems to collection infrastructure should be encrypted using TLS 1.2 or later. Unencrypted data feeds can be intercepted and modified.

- Source authentication — collection agents should authenticate the source of data before ingesting it. Unauthenticated data feeds create an injection point for poisoning attacks.
- Audit logging — every access to the collection pipeline, every configuration change, and every data export should be logged and periodically reviewed.
- Access controls — the training dataset should be accessible to the minimum number of principals required. Broad read access to sensitive training data increases the risk of exfiltration.
- Integrity verification — apply cryptographic hashes to datasets at ingestion and verify those hashes before training begins. Any mismatch indicates potential tampering.
- Network segmentation — data collection infrastructure should be isolated from production security tooling so that a compromise of the AI pipeline does not directly expose production controls.

6. Data Provenance and Chain of Custody

Provenance is the documented history of a piece of data: where it came from, when it was collected, how it was collected, what transformations were applied, and who accessed it. In legal and compliance contexts, provenance provides the audit trail required to demonstrate that data was handled appropriately. In AI security contexts, provenance enables defenders to trace a model's behaviour back to specific training examples — which is essential when investigating unexpected or incorrect predictions.

Practical provenance controls include tagging every record with its source system identifier, collection timestamp, and collection method at ingestion time. When data is transformed (normalised, filtered, resampled), each transformation should be logged with the parameters used. Model training runs should record which dataset version was used so that any model version can be associated with a specific, retrievable training corpus. Tools such as DVC (Data Version Control), Apache Atlas, or custom metadata databases support provenance tracking at scale.

7. Handling Sensitive Data Responsibly

Security datasets are among the most sensitive in any organisation. They may contain personally identifiable information (usernames, IP addresses, email addresses, device identifiers), confidential details about the organisation's security posture (vulnerability status, detection rules, response playbooks), and in some cases regulated categories of information (healthcare records, financial transactions, government data).

The principle of data minimisation — collecting only the information strictly necessary for the AI objective — is both a regulatory requirement under frameworks like GDPR and a security best practice. More data means more exposure if the dataset is compromised. Anonymisation and pseudonymisation techniques reduce privacy risk: replacing usernames with consistent hashed identifiers preserves the behavioural pattern needed for UBA models without retaining the actual identity. However, re-identification risk must be assessed — anonymised datasets can sometimes be re-identified by combining fields, so de-identification should be validated by someone familiar with the specific dataset.

8. Labelling and Ground Truth

Supervised learning requires labelled examples — records where the correct answer is already known. In cybersecurity, creating high-quality labels is expensive because most cases require expert judgment. Is this network connection a beacon or legitimate software update behaviour? Is this login anomaly an insider threat or a new device? Is this file a previously unseen malware variant or a legitimate application with unusual characteristics?

Ground truth is created through several methods: historical incident data (records of confirmed attacks with their associated artefacts), expert analyst labelling (security professionals manually classifying records), sandbox execution (running files in isolated environments to observe malicious behaviour), and threat intelligence cross-referencing (matching indicators against known-good and known-bad repositories). Each method has trade-offs between scale, accuracy, and cost. Expert labelling is most accurate but least scalable. Automated labelling scales well but introduces label noise. Production AI systems typically combine both: automated labelling for high-confidence cases, expert review for ambiguous ones.

Key Risk: Inaccurate labels are silent — the model trains without error but learns incorrect patterns. Label quality audits (sampling and reviewing a random subset of labels) should be a standard part of any data preparation workflow.

9. Balanced Data Collection

Real-world security data is severely imbalanced. On a typical enterprise network, malicious events may represent fewer than 0.1 percent of all logged activity. If a model trains on raw production data without addressing this imbalance, it will learn that the correct answer is almost always 'benign' — and achieve high overall accuracy while failing catastrophically at its actual job of detecting attacks.

Addressing imbalance begins during collection. Where possible, deliberately over-collect examples of the minority class (attacks, anomalies, malicious files) relative to the majority class. This may involve sourcing additional samples from threat intelligence repositories, public malware repositories (VirusTotal, MalwareBazaar), or academic security datasets. Where additional real examples are unavailable, carefully generated synthetic data using techniques such as SMOTE (Synthetic Minority Over-sampling Technique) can supplement the training set, provided that synthetic examples are validated by domain experts before use.

10. Temporal Considerations

Cybersecurity is a continuously evolving domain. Attack techniques, threat actor tooling, and defensive technology all change rapidly. A model trained on data from two years ago may be highly accurate against historical threats while being blind to current ones. This temporal decay problem is distinct from concept drift in other domains because the adversary is actively working to make old detection signatures obsolete.

Training data should include recent examples that reflect the current threat environment. For time-series data such as user behaviour or network traffic, the training period should cover sufficient variation to capture business rhythms: normal business hours versus overnight periods, weekday patterns versus

weekends, end-of-quarter activity spikes, and any major organisational changes (mergers, office moves, system migrations) that shifted baseline behaviour. When evaluating a model, use temporal splitting — training on older data and testing on more recent data — rather than random splitting, because random splitting allows the model to 'see the future' during training, producing unrealistically optimistic evaluation metrics.

11. Legal and Ethical Data Collection

Security teams sometimes operate under the assumption that collecting internal data for defensive purposes is unambiguously permissible. In practice, this assumption is not always accurate. Employment contracts, privacy policies published to employees, data protection regulations, and cross-border data transfer rules may all impose constraints on what employee data can be collected, how it can be stored, and what it can be used for.

- Obtain approval — before collecting personal data for AI training, verify that the organisation's legal team has assessed and approved the collection for this specific purpose.
- Document purpose limitation — data collected for security monitoring may not be repurposable for AI training without additional legal basis, particularly under GDPR's purpose limitation principle.
- Cross-border constraints — data sovereignty laws in some jurisdictions (EU, China, Russia) restrict transferring certain categories of data across national borders for processing.
- Retention limits — data collected for security purposes often has mandated retention windows; training datasets should not be retained beyond what is legally permissible.
- Employee notification — in many jurisdictions, employees must be informed that their activity may be monitored and that the data may be used to train AI models.

12. Data Collection Infrastructure and Documentation

At enterprise scale, security environments generate millions of events daily. Manual collection is impractical; organisations need automated, scalable infrastructure that ingests data reliably, validates it at ingestion, routes it to appropriate storage, and monitors pipeline health. Modern data pipelines for security AI typically include message queues (Apache Kafka, AWS Kinesis) for high-throughput event streaming, distributed object storage (AWS S3, Azure Blob) for raw data archiving, stream processing frameworks (Apache Flink, Spark Streaming) for real-time normalisation, and monitoring dashboards that alert on pipeline failures or data quality anomalies.

Documentation is as important as the infrastructure itself. Every data collection pipeline should document: what data is collected, which source systems feed it, how frequently it is updated, what transformations are applied at ingestion, who owns and maintains the pipeline, what quality checks are applied, what known limitations or gaps exist, and how access is controlled. Well-documented pipelines are easier to audit, easier to hand over between team members, and easier to adapt when source system formats change — which they will.

Key Terms Glossary

Term	Definition
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Ground truth	The correct label for a training example, determined by an authoritative source such as expert analysis, sandbox execution, or confirmed incident data, used to teach a supervised model what it should predict.
Data poisoning	An adversarial attack in which false or misleading records are deliberately injected into a training dataset to corrupt model behaviour, such as causing the model to ignore a specific attack technique.
Data provenance	The documented history of a dataset, including its origin, collection method, collection time, processing steps, and chain of custody, used to verify integrity and support audit requirements.
Data minimisation	The practice of collecting only the data strictly necessary for a defined purpose, reducing privacy risk and limiting exposure in the event of a dataset compromise.
SMOTE	Synthetic Minority Over-sampling Technique — an algorithm that generates synthetic training examples for an underrepresented class by interpolating between existing minority-class instances.
Temporal splitting	A data splitting strategy in which training data is drawn from an earlier time period than test data, simulating real deployment where the model must generalise to future, unseen events.
Concept drift	The phenomenon in which the statistical properties of the target variable change over time, causing a model trained on historical data to become less accurate as the real-world environment evolves.
Pseudonymisation	A data de-identification technique that replaces directly identifying fields (usernames, email addresses) with consistent pseudonyms (hashed identifiers), preserving analytical patterns while reducing privacy exposure.
Label noise	Inaccuracies in training labels caused by misclassification, ambiguous cases, or automated labelling errors, which reduce model quality by teaching incorrect associations.
Data sovereignty	Legal requirements that data collected within a jurisdiction must be stored and processed within that jurisdiction, restricting cross-border data transfers for AI training.

Quick Revision Checklist

- Data quality is the foundation of AI security effectiveness — poor data produces unreliable models regardless of algorithmic sophistication.
- Define data requirements before collection begins by asking what signals an expert analyst would examine to make the same decision the AI will make.
- Security data sources include SIEM logs, firewall records, endpoint telemetry, cloud API logs, identity provider logs, DNS records, and external threat intelligence feeds.
- Evaluate trustworthiness on four dimensions: accuracy, completeness, integrity, and provenance — all four must be addressed.

- Data poisoning is an active adversarial threat; defend against it with access controls, cryptographic hash verification, anomaly detection on ingestion data, and independent validation sets.
- Secure the collection pipeline: encrypt data in transit, authenticate sources, maintain audit logs, apply least-privilege access, and segment the AI infrastructure from production controls.
- Maintain data provenance: tag every record with source, collection time, and method; log all transformations; link model training runs to specific dataset versions.
- Apply data minimisation and pseudonymisation for personal data; verify de-identification is robust against re-identification attacks.
- High-quality ground truth requires expert analyst involvement — automated labelling at scale must be audited with regular label quality reviews.
- Address class imbalance during collection by deliberately over-collecting minority-class examples (attacks); supplement with validated synthetic data where real examples are scarce.
- Use temporal splitting for evaluation, not random splitting, to avoid artificially optimistic metrics that do not reflect real deployment performance.
- Verify legal and regulatory compliance before collecting personal employee data for AI training — purpose limitation, cross-border transfer rules, and employee notification requirements all apply.
- Data collection is a continuous capability, not a one-time event — living datasets that grow and adapt alongside evolving threats are essential for maintaining model relevance.

10 Exam-Based MCQs — Data Collection for AI

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: A security team is building a malware classification model. The team has just been offered access to a new external threat intelligence feed from a vendor they have not previously used. The feed claims to contain 50,000 labelled malware samples. What action should the data engineering team take FIRST before accepting the new external threat feed?

- A) Immediately ingest the feed into the training pipeline to benefit from the additional threat intelligence
- B) Verify the feed's provenance, check its integrity using cryptographic hashes, and assess the vendor's security practices before ingestion
- C) Run the feed through the existing model to see whether it improves detection rates before deciding to accept it
- D) Apply SMOTE to the feed data to balance it with the existing training dataset

Answer: B **Explanation:** B is correct because external data sources must be verified for trustworthiness — provenance, integrity, and vendor security — before being incorporated into a training pipeline. An unverified external feed is a potential data poisoning vector. A is wrong because immediate ingestion of unverified external data is the exact scenario that data poisoning defences are designed to prevent. C is wrong because testing against the current model does not verify the feed's integrity or the vendor's practices. D is wrong because SMOTE addresses class imbalance, not data trustworthiness.

Q2. Scenario: A security data scientist has trained a network intrusion detection model on 18 months of historical NetFlow data. The training and validation metrics look excellent, but after deployment, the false negative rate in production is significantly higher than the validation results suggested. The model's evaluation metrics appear excellent, but production performance is significantly worse than expected. Which data collection practice is MOST likely responsible?

- A) The training dataset was too large, causing the model to overfit to rare edge cases
- B) Random splitting was used for evaluation, allowing the model to learn from future data during training and producing artificially optimistic metrics
- C) The training dataset included too many minority-class examples, biasing the model toward detecting attacks
- D) The data was encrypted in transit, which degraded the quality of the features available for training

Answer: B **Explanation:** B is correct because random splitting in time-series security data allows records from later periods to appear in the training set, meaning the model has 'seen the future' during training. This produces optimistic evaluation metrics that do not reflect how the model will perform when it must generalise to genuinely unseen future data. Temporal splitting — training on older data, testing on more recent data — prevents this. A is wrong because large datasets generally improve generalisation, not reduce it, and overfitting to rare examples would produce high minority-class training accuracy, not better overall metrics. C is wrong because including adequate minority-class examples is necessary for good detection performance. D is wrong because encryption in transit protects data during transfer and does not affect feature quality in the training dataset.

Q3. Scenario: A threat intelligence analyst suspects that an attacker may be attempting to manipulate the organisation's AI-based intrusion detection system by corrupting its training data. Which control BEST protects against a data poisoning attack targeting the AI model's training pipeline?

- A) Using a larger training dataset so that a small number of poisoned records have less statistical influence
- B) Encrypting the trained model weights so that attackers cannot reverse-engineer them
- C) Applying cryptographic hashes to datasets at ingestion and verifying those hashes before each training run
- D) Enabling multi-factor authentication for analyst accounts that access the production model

Answer: C **Explanation:** C is correct because cryptographic hash verification detects any modification to the dataset between the time it was collected and the time it is used for training. If an attacker has inserted poisoned records, the hash will not match the expected value and the training run will be blocked pending investigation. A is partially true — larger datasets dilute the influence of a small number of poisoned records — but it is not a control; it is a mitigating factor that reduces impact without detecting the attack. B is wrong because encrypting model weights protects the model from theft or analysis, not the training data from poisoning. D is wrong because MFA protects analyst accounts but does not prevent an attacker who has already compromised an account or the pipeline itself from inserting poisoned records.

Q4. Which of the following BEST describes the purpose of data provenance in AI security?

- A) It ensures that the training dataset is sufficiently large to support deep learning model architectures
- B) It provides a documented history of a dataset's origin, collection method, and transformations, enabling integrity verification and audit
- C) It anonymises personal data in the training set to comply with privacy regulations
- D) It balances the distribution of classes in the training dataset to improve minority-class detection

Answer: B **Explanation:** B is correct because provenance is the documentation chain that answers 'where did this data come from and how has it been handled?' — enabling organisations to verify integrity, trace model behaviour to specific training examples, and satisfy audit requirements. A is wrong because dataset size is a volume concern, not a provenance concern. C is wrong because anonymisation is a privacy control, distinct from provenance tracking (though both may be implemented in the same pipeline). D is wrong because class balancing is an imbalance-handling technique, not a provenance concern.

Q5. A security team is building a user behaviour analytics (UBA) model that will process employee authentication logs. Under GDPR, which requirement is MOST directly relevant to this data collection activity?

- A) The model must achieve at least 95 percent accuracy before personal data can be used for training
- B) The organisation must have a legal basis for processing personal data, and data collected for security monitoring may not be automatically repurposable for AI training
- C) Personal data must be encrypted using AES-256 before it can be included in any AI training dataset
- D) The model must be able to explain every prediction made on personal data before training can begin

Answer: B **Explanation:** B is correct because GDPR's purpose limitation principle requires that data collected for one purpose (security monitoring) cannot be automatically repurposed for another (AI model training) without a separate legal basis. Organisations must assess whether AI training is covered by their existing legal basis or whether a new basis (such as legitimate interests) must be established. A is wrong because GDPR does not set accuracy thresholds for AI models. C is wrong because while encryption is a best practice, GDPR's primary requirement is lawful processing, not a specific encryption standard. D is wrong because GDPR's explainability requirements (Article 22) relate to automated decisions affecting individuals, not to the training process itself.

Q6. Which of the following data collection practices MOST effectively addresses class imbalance in a security AI training dataset?

- A) Collecting a larger total volume of data from production logs to increase the dataset size
- B) Deliberately over-collecting minority-class examples (malicious events) from threat repositories and supplementing with validated synthetic samples
- C) Removing duplicate records from the majority class to reduce its relative proportion
- D) Applying data encryption to majority-class records to reduce their influence during training

Answer: B **Explanation:** B is correct because addressing imbalance requires increasing the representation of the minority class through targeted collection of additional malicious examples from threat intelligence repositories, public malware datasets, or validated synthetic generation. A is wrong because collecting more data from production logs proportionally increases both classes — a 99:1 imbalance on a million records is still 99:1 on ten million records. C describes undersampling the majority class, which is one valid technique, but B describes a more complete approach that increases minority-class diversity rather than just discarding majority-class data. D is wrong because encryption affects data confidentiality, not class distribution or training dynamics.

Q7. What is the MOST significant risk of creating AI training labels through automated rule matching alone, without expert review?

- A) Automated labelling is slower than expert labelling, increasing project timelines
- B) Automated labelling produces labels in machine-readable formats that some ML frameworks cannot ingest
- C) Automated labelling introduces label noise when rules misclassify ambiguous or novel cases, teaching the model incorrect patterns
- D) Automated labelling creates labels that are too detailed, causing the model to overfit to specific syntax rather than learning general patterns

Answer: C **Explanation:** C is correct because automated rules are designed for known patterns and fail on edge cases, ambiguous records, or novel attack techniques. When these records are incorrectly labelled and included in training data, the model learns wrong associations — a problem called label noise. Label noise degrades model quality silently because no training error is raised; the model simply learns incorrect patterns. A is wrong because automated labelling is generally faster than expert labelling, not slower. B is wrong because label format compatibility is a data engineering concern, not a quality risk. D is wrong because label granularity does not cause overfitting — model architecture and regularisation are the relevant controls.

Q8. Which security control is MOST important to apply to a data collection pipeline that ingests sensitive endpoint telemetry?

- A) Storing all endpoint telemetry in plaintext for faster query performance during model training
- B) Encrypting data in transit using TLS and authenticating source agents before ingesting telemetry
- C) Publishing the pipeline's data schema publicly to allow external researchers to contribute additional training samples
- D) Routing all endpoint telemetry through the production SIEM before forwarding it to the AI pipeline

Answer: B **Explanation:** B is correct because the two most critical controls for a data collection pipeline handling sensitive telemetry are: encrypting data in transit (to prevent interception and modification) and authenticating sources (to prevent injection of false data by an attacker impersonating a legitimate endpoint). Together, these controls address confidentiality and integrity of the data pipeline. A is wrong because storing sensitive telemetry in plaintext is a fundamental confidentiality failure. C is wrong because publishing the schema publicly creates an attack surface for data poisoning and exposes information about

the organisation's detection capabilities. D is wrong because routing through the production SIEM introduces unnecessary coupling between the operational and AI infrastructure.

Q9. What does the principle of data minimisation require when collecting employee activity logs for an AI insider-threat detection model?

- A) Collecting the maximum amount of employee data available to give the AI model the best possible feature set
- B) Collecting only the data strictly necessary for the insider-threat detection objective, and no more
- C) Minimising the size of each individual log record by truncating fields to their shortest useful length
- D) Collecting data from the minimum number of source systems to reduce infrastructure complexity

Answer: B **Explanation:** B is correct because data minimisation — a core principle in GDPR and broader data protection frameworks — requires limiting collection to data that is strictly necessary for the defined purpose. For insider-threat detection, this means collecting access logs, data movement records, and authentication data, but not unrelated personal data such as personal emails, health records, or off-hours device usage if those signals are not required by the model. A is wrong because collecting the maximum available data violates data minimisation principles and increases privacy and security risk. C is wrong because minimisation refers to the scope of data collected (which fields, which individuals) not to record field length. D is wrong because source system count is an infrastructure concern, not a data minimisation decision.

Q10. An organisation's malware classification model was performing well six months ago but is now missing a significant percentage of new malware samples in production. Which data collection issue MOST likely explains this degradation?

- A) The training dataset contained too many minority-class examples, causing the model to over-detect malware
- B) The training data did not include recent samples reflecting current malware techniques, resulting in concept drift
- C) The collection pipeline is applying too much encryption, degrading the features available for inference
- D) The organisation is collecting too much network traffic data, overwhelming the model with majority-class examples

Answer: B **Explanation:** B is correct because cybersecurity is adversarial — malware authors continuously evolve their techniques to evade detection. A model trained on data from six months ago may have learned to recognise historical techniques that are now obsolete while being blind to new ones. This is concept drift: the statistical properties of the target (malware behaviour) have changed, making the historical training data less representative of current production inputs. Continuous data collection with regular retraining on fresh samples is the primary remedy. A is wrong because over-detection (too many false positives) is the opposite of the described problem (missing malware = false negatives). C is wrong because encryption of data in transit does not affect the features available to a model during inference. D is

wrong because a high volume of benign traffic would increase false negatives only if the model had not been trained with appropriate class balancing — and this would have been a problem from deployment, not a degradation over time.